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Forest biomass carbon stock estimates via a novel approach: K-nearest neighbor-based weighted least squares multiple birth support vector regression coupled with whale optimization algorithm

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ABSTRACT

Multiple birth support vector regression (MBSVR) provides fast computation and superior performance but overlooks local sample information and has challenges in parameter selection. The traditional least squares models boast fast computational speed but lack robustness and may struggle with noise and outliers. The carbon storage estimates are easily affected by noise and interference points. MBSVR and least squares models are only partially effective in carbon storage estimates. Consequently, we propose least squares multiple birth support vector regression (LSMBSVR) and K-nearest neighbor-based (KNN) weighted least squares multiple birth support vector regression (WLSMBSVR), which have the following merits. Firstly, both models inherit the strengths of MBSVR. Secondly, they exhibit enhanced fitting accuracy, robust stability, and remarkable anti-interference capability. Thirdly, LSMBSVR offers a faster training speed and maintains a comparable regression performance to MBSVR. Fourthly, WLSMBSVR considers the local information, enhancing its anti-interference capability. Lastly, we employ the whale optimization algorithm (WOA) to improve the effectiveness of parameter selection. Experiment results indicate that our models can be more effective on carbon storage, synthetic, and UCI datasets than compared models, verifying the broad application value of our models.

1. Introduction

Forest carbon stock forecasting is inherently vulnerable to a range of disruptive factors. It necessitates careful management of noise and interference. Firstly, undocumented natural events like wildfires, pest outbreaks and as well as human-induced activities such as deforestation and land conversion, are significant sources of predicted disturbances (Valentini et al., 2000). These factors can disrupt carbon retention and cause long-term changes to forest structure and growth, ultimately affecting the capacity of forests for carbon absorption. Secondly, the measurement process is subject to noise. The limitations of point sampling potentially lead to an incomplete grasp of forest carbon dynamics (Chimner et al., 2023). Thirdly, carbon storage data can be influenced by a variety of factors, including forest type, age, soil type, climate conditions, human activities, etc. (Johnson et al., 2023). The diversity of data collection methods and measurement errors often lead to outliers or noise, compromising the accuracy and reliability of the data (Hill et al., 2012). Fourthly, these studies are marred by significant uncertainties stemming from an absence of comprehensive ground-based data (Pan et al., 2011). Lastly, there is inconsistency in the timing and methodologies of data collection (Ni, 2013). These factors underscore the need for more advanced analytical techniques to mitigate the impact of noise and interference in forest carbon stock estimates. Then, we can more accurately assess the role of forests as carbon sinks and provide a scientific foundation for developing effective forest management and climate change mitigation strategies.

In the field of data-driven sustainability research, studies have highlighted the key role of data-driven technologies in achieving sustainable development goals. The research shows that environmental monitoring and carbon management can be effectively conducted using data-driven methods, providing a scientific basis for ecological decision-making (Mohammed et al., 2023). The accuracy of carbon stock estimation is mainly dependent on the techniques and algorithms employed. The ordinary least squares (OLS) method has been a fundamental model in ecological research (Dube and Mutanga, 2015). However, OLS is often inadequate for handling complex regression problems. Especially in

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carbon storage estimation, there are issues with high-dimensional nonlinearity and feature correlation. To address these limitations, machine learning algorithms, including random forests (RF), support vector machines (SVM), and extreme gradient boosting (XGBoost), have been developed to manage multi-source data and to research forest ecosystems effectively (Zhu et al., 2024; Sabat-Tomala et al., 2024; Jia et al., 2024).

SVM is one of the very popular algorithms in machine learning (Vapnik, 1995). The SVM is adept at handling high-dimensional nonlinear regression problems due to its kernel-based approach. Its structure risk minimization property endows the model with enhanced robustness. It classifies samples by finding the optimal hyperplane that maximizes margin, relying on support vectors (Fang et al., 2021). Support vector regression (SVR) is the application of SVM in regression problems. SVR aims to find a hyperplane that minimizes prediction error while maximizing the margin to ensure strong generalization capabilities (Smola and Schölkopf, 1998). Compared to traditional regression algorithms such as linear regression, polynomial regression, ridge regression, etc, SVR excels at capturing complex nonlinear relationships, ensures robust generalization through structural risk minimization, and demonstrates enhanced tolerance to noise (Liu et al., 2021). Due to these strengths, SVR has found widespread applications in various domains, including medical data analysis, agricultural development, engineering prediction and ecological conservation (Park et al., 2024; Ganaie et al., 2023; Fu et al., 2023; Chen et al., 2023).

However, traditional SVR has these limitations: (1) It exhibits high time complexity, resulting in slow computation speed (GhaffariRazin et al., 2023). (2) It lacks robustness in dealing with noise and outliers and pays insufficient attention to local information among samples (Gu et al., 2020). (3) Its parameter adjustment is complex, requiring simultaneous adjustment of multiple key parameters, making it sensitive to data and difficult to balance complexity and performance (Nguyen et al., 2021). These limitations have motivated researchers to develop improved SVR models.

In terms of high time complexity and slow computation speed, numerous improved models have been proposed to address these problems. One of the most common SVR models, ε-support vector regression (ε-SVR), is designed to accommodate all data points above and below the ε-tube on the hyperplane. It uses an ε-insensitive loss function, treating data points within the ε-tube as satisfying fitting requirements (Bi and Bennett, 2003). Twin support vector regression (TSVR) addresses two small-scale quadratic programming problems (QPPs), improving the speed of the original SVR. Additionally, this method also shows improved performance for simultaneously solving multiple subproblems (Peng, 2010). Least squares support vector regression (LSSVR) has been recognized for employing inequality constraints rather than equality constraints, thereby transforming the problem into an optimization of the Lagrangian function, which is ultimately reduced to solving a system of linear equations. This transformation significantly enhances the training speed of the model. However, research has indicated that it lacks robustness and often yields mediocre performance in regression tasks (Zhao et al., 2013). The least squares twin support vector regression (LSTSVR) was initially proposed by Huang et al. which can be seen as an improvement of TSVR (Huang et al., 2013). While employing the L_2 norm offers computational convenience, least squares models may result in a less efficient algorithmic performance due to the consequent reduction in sparsity. This approach complicates the feature selection process and augments the vulnerability to outliers. Concurrently, LSTSVR incurs redundancy in its terms, which can further impede its efficiency and accuracy (Huang et al., 2022; Li et al., 2023). In 2021, Zhang et al. introduced multiple birth support vector regression (MBSVR), with improved speed and performance (Zhang et al., 2021). In MBSVR, data is partitioned into several classes, and during each iteration of solving QPPs, only one of the classes actively participates in the final iteration. It results in fewer iterations, leading to a significant time improvement. Experiments demonstrate its feasibility, superior fitting, and better regression performance. However, these models overlook local sample information and have challenges in parameter selection. Firstly, there are often some subtle local changes in real data, which may contain important patterns or trends. Models that ignore local information cannot capture the rules and characteristics. Especially when there are some noise and interference points in the data, these points may significantly impact the fitting of the model. Secondly, when the model involves multiple parameters, we often need to spend much time and effort on parameter debugging and optimization, trying different parameter combinations to find the best parameter settings. This process is not only time-consuming and labor-intensive, but also difficult to find the global optimal solution because of the large parameter space.

In terms of overlooking local information among samples, these models above have difficulty handling noise and outliers (Gu et al., 2020). There are different strategies to enhance the consideration of local information, such as clustering weighting, feature selection weighting, and so on. In this way, the model can not only deal with the global data trend but also capture the local subtle changes, so it can better identify and utilize the local structural information in the face of complex data environment. Subsequently, numerous improved models have been proposed, including weighted TSVR (Xu and Wang, 2012), fast clustering-based weighted TSVR (FC-WTSVR) (Gu et al., 2020), projection weighted TSVR (PWWTSVR) (Wang et al., 2021), dynamic weight distribution strategy SVR (SVR-ADWD) (Xing et al., 2024), and feature-weighted SVR (Zhang et al., 2024). These models give more significant consideration to the overall structure of the samples. Consequently, they exhibit enhanced fitting performance and generalization capabilities when dealing with complex noise and outlier interference (Wang et al., 2023). The application of the KNN algorithm is prevalent across various integrations and applications (Zhang and Li, 2023; Wang et al., 2022).

Regarding parameter selection, the tuning process of the models above requires a careful balance between complexity and performance. Traditional grid search (GS) requires discretization of the parameter space into a grid, which may result in the accuracy of the solution being limited by the mesh granularity and may miss a better solution on non-grid points. Heuristic algorithms do not rely on discretization of the parameter space and can be optimized in a continuous parameter space. Many heuristic algorithms are combined with machine learning models to solve parameter-tuning problems. Chen et al. proposed SVR combined particle swarm optimization (PSO) to predict the carbon emissions level in China's logistics industry (Chen et al., 2024). Han et al. researched the deformation uniformity control of thin-walled conical aeroengine forgings based on SVR combined genetic algorithms (GA) (Han et al., 2024). Karasu and Saraç combined 2D Riesz transform, multi-objective gray wolf optimizer (MOGWO), and KNN to study the classification of power quality interference (Karasu and Saraç, 2020). Mirjalili and Lewis introduced the WOA (Mirjalili and Lewis, 2016), which simulates the migration of whales and foraging processes. It optimizes the objective function by iteratively updating positions, characterized by its global search capabilities and rapid convergence, making it suitable for complex optimization problems. WOA has been combined with various models to achieve promising results. For instance, a hybrid SVR-WOA model was proposed to apply in the agricultural sector (Nguyen et al., 2021), and 2D discrete wavelet transform combined with WOA-KNN classifier was proposed to study the automatic recognition and classification of power quality interference (Karasu and Saraç, 2022).

In conclusion, despite numerous models introduced to overcome the restrictions of traditional SVR, a comprehensive method that effectively tackles the shortcomings of SVR still needs to be improved. Moreover, a model that can efficiently address the individual inadequacies of the MBSVR and least squares models is needed to accomplish the challenges of forest carbon stock estimates. To this end, we propose two novel models: LSMBSVR and WLSMBSVR. LSMBSVR offers the following advantages:

Table 1
Comprehensive information of carbon storage datasets.

Dataset	Response variable	Independent variables	Data
ORCd	Cd	37	1369
ORDH	DH	35	1369
ORD	D	35	1369

(1) LSMBSVR inherits the advantages of MBSVR, such as feasibility, superior fitting, and better regression performance.
(2) By leveraging the least squares method, LSMBSVR transforms the process of solving QPPs into solving a system of linear equations. This significantly improves the training speed compared to MBSVR.
(3) LSMBSVR avoids redundancy in its terms. While LSTSVR consists of a two-term loss function, LSMBSVR is composed of a regularization term and loss functions. Therefore, LSMBSVR has better computational efficiency and accuracy.

To further enhance the handling of noise and outliers, we mine the local information among samples by using KNN. Building upon LSMBSVR, we propose WLSMBSVR, which inherits the outstanding performance of LSMBSVR. The tuning process of these models above requires a careful balance between complexity and performance. The traditional method GS has the problems of time-consuming and insufficient accuracy. To overcome parameter tuning challenges, we further integrate WOA with our models, enhancing their effectiveness.

Our contributions are as follows: To address the challenges posed by disturbed points in carbon storage estimation, we propose LSMBSVR and WLSMBSVR models combined with WOA and compare them with SVR, TSVR, LSTSVR, and MBSVR. In addition, to further compare the model performance, we conduct experiments on carbon storage, UCI, and synthetic datasets. The results demonstrate that LSMBSVR achieves a faster solving speed, while WLSMBSVR exhibits superior anti-interference and generalization ability. To verify the effectiveness of WOA, we compare GS-based parameter selection and WOA-based parameter selection across all six models.

The remaining sections of this paper are organized as follows: Section 2 is the material information and related works. Section 3 is the theoretical derivation of our models LSMBSVR, WLSMBSVR, and the WOA typed algorithm. Section 4 is the analysis of the models. Section 5 compares the performance of six models on carbon storage datasets. Section 6 compares the models on synthetic and UCI datasets. Section 7 discusses the models. Section 8 is the conclusion and future works.

2. Material and related works

2.1. Material

The data employed in this experiment are sourced from Luo et al. (2020). The data encompasses forested regions in Northeast Asia. This data of carbon stock estimates includes 43 ecological, remote sensing, and environmental variables. These predictors are categorized into five groups: ecosystem (4 variables), remote sensing (14 variables), bioclimatic (21 variables), anthropogenic (4 variables), and geographic coordinates (longitude and latitude).

In our experimental framework, we perform regression analyses on the following variables: Cd: Forest biomass carbon density (total woody carbon per hectare, measured in Mg C per ha); DH: Dominant tree height (measured in meters); D: Mean diameter at breast height (measured in centimeters) (see Table 1).

2.2. Related works

This section provides the fundamental theories of SVR, TSVR, LSTSVR and MBSVR. Throughout this paper, the symbols used are defined in Table 2.

Table 2
List of notations.

Symbol	Descriptions
(x_i, y_i)	ith training point
l, l_k	Number of all training points and A_k
$\kappa(a, b)$	Kernel function
K	The number of hyperplanes
$L_k, C_k, C_{k_1}, C_{k_2}$	Non-negative penalty parameter
N	The number of K-nearest neighbors
B_k, A_k	Training set of A class and B class
Y_{k_1}, Y_{k_2}	Corresponding values of B_k and A_k
$\|\cdot\|$	2-norm
w_k	Weight vectors for each hyperplane
$\top$	Transpose of matrix or vector
L	Lagrangian function

2.2.1. SVR

SVR aims to find a regression hyperplane represented as: $f(x) = w^\top x + b$, where $x \in \mathbb{R}^m$, $w \in \mathbb{R}^m$, $b \in \mathbb{R}$. Let X denote the training dataset, and Y represent the corresponding values, with a total of l data points and m features, where $X \in \mathbb{R}^{l\times m}$, $Y \in \mathbb{R}^l$. The optimization problem that SVR aims to solve is formulated as follows:

$$\begin{aligned} \min_{w,b,\xi,\xi^*} \quad & \frac{1}{2}\|w\|^2 + Ce^\top\left(\xi + \xi^*\right) \\ \text{s.t.} \quad & Y - Xw - eb \le e\varepsilon + \xi, \\ & Xw + eb - Y \le e\varepsilon + \xi^*, \\ & \xi \ge 0, \quad \xi^* \ge 0, \end{aligned} \tag{1}$$

where $C > 0$ serves as a regularization parameter, controlling the complexity, ξ and ξ^* represent slack variables, $\varepsilon > 0$ and e is an all-ones vector.

2.2.2. TSVR

TSVR seeks two non-parallel hyperplanes, $f_1(x) = w_1^\top x + b_1$ and $f_2(x) = w_2^\top x + b_2$. By solving two smaller-scale QPPs, eventually, the final regression hyperplane $f(x) = \frac{1}{2}\left(f_1(x) + f_2(x)\right)$ is obtained by averaging these two hyperplanes. The optimization problems are:

$$\begin{aligned} \min_{w_1,b_1,\xi_1} \quad & \frac{1}{2}\left\|Y - e\varepsilon_1 - \left(Xw_1 + eb_1\right)\right\|^2 + C_1e^\top\xi_1 \\ \text{s.t.} \quad & Y - \left(Xw_1 + eb_1\right) \ge e\varepsilon_1 - \xi_1, \\ & \xi_1 \ge 0, \end{aligned} \tag{2}$$

and

$$\begin{aligned} \min_{w_2,b_2,\xi_2} \quad & \frac{1}{2}\left\|Y + e\varepsilon_2 - \left(Xw_2 + eb_2\right)\right\|^2 + C_2e^\top\xi_2 \\ \text{s.t.} \quad & \left(Xw_2 + eb_2\right) - Y \ge e\varepsilon_2 - \xi_2, \\ & \xi_2 \ge 0. \end{aligned} \tag{3}$$

Unlike SVR, which combines upper and lower ε-insensitive bounds in a single QPP, TSVR separates them into two QPPs. The constraints of TSVR are half that of SVR, significantly enhancing the regression speed.

2.2.3. MBSVR

MBSVR divides all training set points $X \in \mathbb{R}^{l\times m}$ into K parts, where $X = [A_1; A_2; \cdots; A_k; \cdots; A_K]$, A_k contains l_k points, where $A_k \in \mathbb{R}^{l_k\times m}$, and B_k represents the complement of A_k on X, where $B_k = [A_1; A_2; \cdots; A_{k-1}; A_{k+1}; \cdots; A_K]$, $B_k \in \mathbb{R}^{(l-l_k)\times m}$, $k = 1, 2, \ldots, K$. The response vectors of X, A_k, and B_k are denoted as $Y \in \mathbb{R}^l$, $Y_{k_2} \in \mathbb{R}^{l_k\times m}$ and $Y_{k_1} \in \mathbb{R}^{(l-l_k)\times m}$ (Zhang et al., 2021). For each different QPP, two sets can be constructed:

$$D^+ = \left\{(A_k, g)\right\}, \quad D^- = \left\{(B_k, f)\right\}, \tag{4}$$

where $f = Y_{k_1} - e_{k_1}\varepsilon_k$, $g = Y_{k_2} + e_{k_2}\varepsilon_k$. The optimization problem for each MBSVR is defined as:

$$\begin{aligned} \min_{w_k,b_k,\xi_k} \quad & \frac{L_k}{2}\left(w_k^2 + b_k^2\right) + \frac{1}{2}\left(B_k w_k + e_{k_1} b_k - \left(Y_{k_1} - e_{k_1}\varepsilon_k\right)\right)^2 \\ & + C_k e_{k_2}^{\top}\xi_k \\ \text{s.t.} \quad & Y_{k_2} + e_{k_2}\varepsilon_k - \left(A_k w_k + e_{k_2} b_k\right) \le \xi_k, \end{aligned} \tag{5}$$

where $e_{k_1} \in \mathbb{R}^{(l-l_k)\times m}$, $e_{k_2} \in \mathbb{R}^{l_k \times m}$, $C_k \in \mathbb{R}^l > 0$, ξ_k is slack variable, $\xi_k \in \mathbb{R}^{l_k}$, $\xi_k > 0$ and $L_k > 0$. In a iterative process where k ranges $k = 1, 2, \ldots, K$, we obtain K hyperplanes, then the final regressor is:

$$f(x) = \frac{1}{K}\sum_{k=1}^{K} f_k(x) = \frac{1}{K}\sum_{k=1}^{K}\left({w_k}^{\top} x + b_k\right). \tag{6}$$

Notely, ξ_k only minimizes the fitting error for class A points, thereby less number of points in the constraints of QPPs, which enhances the solution efficiency.

2.2.4. LSTSVR

LSTSVR is initially proposed by Huang et al. the least squares form of TSVR (Huang et al., 2013). The formula is as follows:

$$\begin{aligned} \min_{w_1,b_1,\xi_1} \quad & \frac{1}{2}\left\|Y - e\varepsilon_1 - \left(Xw_1 + eb_1\right)\right\|^2 + \frac{1}{2}C_1\xi_1^T\xi_1 \\ \text{s.t.} \quad & Y - \left(Xw_1 + eb_1\right) = -e\varepsilon_1 - \xi_1, \end{aligned} \tag{7}$$

and

$$\begin{aligned} \min_{w_2,b_2,\xi_2} \quad & \frac{1}{2}\left\|Y + e\varepsilon_2 - \left(Xw_2 + eb_2\right)\right\|^2 + \frac{1}{2}C_2\xi_2^T\xi_2 \\ \text{s.t.} \quad & \left(Xw_2 + eb_2\right) - Y = -e\varepsilon_2 - \xi_2. \end{aligned} \tag{8}$$

While the employment of the L_2 norm offers computational convenience, least squares models may result in a less efficient performance due to the reduction in sparsity. This approach complicates the feature selection and augments the vulnerability to outliers. Notely, LSTSVR incurs redundancy at its first and second items, which can further impede its efficiency and accuracy (Huang et al., 2022; Li et al., 2023).

3. Our works

Carbon stock estimates face challenges due to the diversity of data collection, unrecorded data, and measurement errors (Chen et al., 2016; Hill et al., 2012), which can introduce noise and interference. MBSVR provides rapid computation, superior regression capabilities, and robustness. However, it struggles to handle noise and outliers effectively, and faces challenges in parameter selection. These make it less effective for carbon storage estimates. To solve these limitations, we propose two novel algorithms: LSMBSVR and WLSMBSVR, both integrated with WOA.

3.1. LSMBSVR

For the original MBSVR using the square of the L_2 norm of slack variables, we formulate the optimization problem for LSMBSVR:

$$\begin{aligned} \min_{w_k,b_k,\xi_k} \quad & \frac{L_k}{2}\left(w_k^2 + b_k^2\right) + \frac{1}{2}\left(B_k w_k + e_{k_1} b_k - \left(Y_{k_1} - e_{k_1}\varepsilon_k\right)\right)^2 \\ & + \frac{C_k}{2}\xi_k^{\top}\xi_k \\ \text{s.t.} \quad & Y_{k_2} + e_{k_2}\xi_k - \left(A_k w_k + e_{k_2} b_k\right) = \xi_k. \end{aligned} \tag{9}$$

The following analyses explain the mechanism of LSMBSVR:

- $\frac{L_k}{2}\left(w_k^2 + b_k^2\right)$ regulates model complexity and prevents overfitting by penalizing large weights w_k and the bias term b_k. The parameter L_k controls the degree of regularization, balancing the fidelity to the training data with complexity to unseen data and then influencing generalization performance.
- $\frac{1}{2}\left(B_k w_k + e_{k_1} b_k - \left(Y_{k_1} - e_{k_1}\varepsilon_k\right)\right)^2$ minimizes the distance betwe en class B points and $Y_{k_1} - e_{k_1}\varepsilon_k$.
- $\frac{C_k}{2}\xi_k^{\top}\xi_k$ minimizes the distance between class A points and $Y_{k_2} + e_{k_2}\varepsilon_k$. The use of the L_2 norm provides favorable mathematical properties, facilitating the transformation of the problem into a system of linear equations.

The Lagrangian function of LSMBSVR can be written as:

$$\begin{aligned} L = & \frac{L_k}{2}\left(w_k^2 + b_k^2\right) + \frac{1}{2}\left(B_k w_k + e_{k_1} b_k - \left(Y_{k_1} - e_{k_1}\varepsilon_k\right)\right)^2 \\ & + \frac{C_k}{2}\left(Y_{k_2} + e_{k_2}\varepsilon_k - \left(A_k w_k + e_{k_2} b_k\right)\right)^2. \end{aligned} \tag{10}$$

By computing the partial derivatives concerning w_k and b_k and setting them to zero, the optimal solution is obtained:

$$\begin{aligned} \frac{\partial L}{\partial w_k} = & L_k w_k + B_k^{\top}\left(B_k w_k + e_{k_1} b_k - \left(Y_{k_1} e_{k_1}\varepsilon_k\right)\right) \\ & + C_k(-A_k)^{\top}\left(Y_{k_2} + e_{k_2}\varepsilon_k - \left(A_k w_k + e_{k_2} b_k\right)\right) = 0, \\ \frac{\partial L}{\partial b_k} = & L_k b_k + e_{k_1}^{\top}\left(B_k w_k + e_{k_1} b_k - \left(Y_{k_1} - e_{k_1}\varepsilon_k\right)\right) \\ & + C_k\left(-e_{k_2}\right)^{\top}\left(Y_{k_2} + e_{k_2}\varepsilon_k - \left(A_k w_k + e_{k_2} b_k\right)\right) = 0. \end{aligned} \tag{11}$$

Arranging the equations, we get:

$$\begin{bmatrix} w_k \\ b_k \end{bmatrix} = \left[L_k I + G^{\top}G + C_k H^{\top}H\right]^{-1} \cdot \left[G^{\top} f + C_k H^{\top} g\right], \tag{12}$$

where $G = \begin{bmatrix} B_k, e_{k_1} \end{bmatrix}$, $H = \begin{bmatrix} A_k, e_{k_2} \end{bmatrix}$. Finally, the resulting equation is as (6). For the nonlinear form, G and H can be replaced by the following equation:

$$G = \begin{bmatrix} \kappa(B_k, X^{\top}), e_k \end{bmatrix} \quad \text{and} \quad H = \begin{bmatrix} \kappa(A_k, X^{\top}), e_k \end{bmatrix}. \tag{13}$$

In terms of speed, by introducing the least squares framework, we transform solving QPPs into solving a system of linear equations. This notably accelerates the regression process. In terms of performance, LSMBSVR originates from MBSVR. Theoretically, LSMBSVR has a comparable performance to MBSVR. Note that the difference between the second and third terms of LSMBSVR does not incur redundancy in its terms. The entire training procedure of LSMBSVR is summarized in Algorithm 1.

Algorithm 1 The training procedure for LSMBSVR.

1: **Input**: The training sample set X, Y, parameters K, L_k, C_k, ε;
2: **Output**: The regression function $f(x)$;
3: Divide X evenly into K parts, $X = [A_1; A_2; \cdots; A_k; \cdots; A_K]$, $Y = [Y_{1_1}; Y_{2_1}; \cdots; Y_{k_1}; \cdots; Y_{K_1}]$, **then** obtain the corresponding B_k, Y_{k_2} for each A_k, Y_{k_1};
4: **for** $k \le K$ **do**
  Obtain G and H from (13), according to (12), **then** get w_k and b_k;
  end for
5: According to (6), obtain the function $f(x)$.

Although LSMBSVR is faster than MBSVR and has comparable performance to MBSVR, its reliance on the L_2 norm compromises sparsity and increases sensitivity to outliers, potentially resulting in overfitting. Improvements to LSMBSVR are warranted. To address these issues, we proposed WLSMBSVR.

3.2. WLSMBSVR

3.2.1. K-nearest neighbor method

For all l points, the weight matrix W is defined as follows:

$$W_{i,j} = \begin{cases} 1, & \text{if } x_i \text{ is } k\text{-nearest neighbors of } x_j \\ & \quad \text{or } x_j \text{ is } k\text{-nearest neighbors of } x_i, \\ 0, & \text{otherwise.} \end{cases} \tag{14}$$

$d_i = \sum_{j=1}^{i} W_{ij}$ represents the cumulative weighted degree of a point for $i = 1, 2, \ldots, l$, reflecting the influence of its neighbors. The weighted diagonal matrix is $D = \mathrm{diag}(d_1, d_2, \ldots, d_l)$. The more neighborhoods a point has, the more important it is, and then given a higher weight (Xu and Wang, 2014).

For all sample points, the weighted diagonal matrix is D. The corresponding parts of classes A and B are separated and denoted as D_1 and D_2. The specific formula is as follows:

$$D = \mathrm{diag}(d_1, d_2, \ldots, d_l),\ D_1 = \mathrm{diag}(d_{B_1}, d_{B_2}, \ldots, d_{B_{(l-l_k)}}) \quad \text{and} \quad D_2 = \mathrm{diag}(d_{A_1}, d_{A_2}, \ldots, d_{A_{l_k}}). \tag{15}$$

3.2.2. WLSMBSVR

We use KNN weights to construct WLSMBSVR, and the original problem is as follows:

$$\begin{aligned} \min_{w_k, b_k, \xi_k} \quad & \frac{L_k}{2}\left(w_k^2 + b_k^2\right) + \frac{C_{k_1}}{2}\left(B_k w_k + e_{k_1} b_k - \left(Y_{k_1} - e_{k_1}\varepsilon_k\right)\right)^{\top} \\ & D_1\left(B_k w_k + e_{k_1} b_k - \left(Y_{k_1} - e_{k_1}\varepsilon_k\right)\right) + \frac{C_{k_2}}{2}\xi_k^{\top} D_2 \xi_k \\ \text{s.t.} \quad & Y_{k_2} + e_{k_2}\xi_k - \left(A_k w_k + e_{k_2} b_k\right) = \xi_k. \end{aligned} \tag{16}$$

The second item minimizes the distance between class B points and $(Y_{k_1} - e_{k_1})$ and given KNN weight. The third item minimizes the distance between class A points and $(Y_{k_2} + e_{k_2})$ and given KNN weight. By leveraging the KNN weights, we can assign different importance levels to individual data points. This strategy enhances the capability to capture intricate patterns and improves its robustness against outliers and noise. The Lagrangian function of WLSMBSVR can be written as:

$$\begin{aligned} L = & \frac{L_k}{2}\left(w_k^2 + b_k^2\right) + \frac{C_{k_1}}{2}\left(B_k w_k + e_{k_1} b_k - \left(Y_{k_1} - e_{k_1}\varepsilon_k\right)\right)^{\top} \\ & D_1\left(B_k w_k + e_{k_1} b_k - \left(Y_{k_1} - e_{k_1}\varepsilon_k\right)\right) \\ & + \frac{C_{k_2}}{2}\left(Y_{k_2} + e_{k_2}\varepsilon_k - \left(A_k w_k + e_{k_2} b_k\right)\right)^{\top} \\ & D_2\left(Y_{k_2} + e_{k_2}\varepsilon_k - \left(A_k w_k + e_{k_2} b_k\right)\right). \end{aligned} \tag{17}$$

By computing the partial derivatives concerning w_k and b_k and setting them to zero, the optimal solution is obtained:

$$\begin{aligned} \frac{\partial L}{\partial w_k} = & L_k w_k + C_{k_1} B_k^{\top} D_1\left(B_k w_k + e_{k_1} b_k - \left(Y_{k_1} - e_{k_1}\varepsilon_k\right)\right) \\ & + C_{k_2}\left(-A_k\right)^{\top} D_2\left(Y_{k_2} + e_{k_2}\varepsilon_k - \left(A_k w_k + e_{k_2} b_k\right)\right) \\ = & 0, \end{aligned} \tag{18}$$

$$\begin{aligned} \frac{\partial L}{\partial b_k} = & L_k b_k + C_{k_1} e_{k_1}^{\top} D_1\left(B_k w_k + e_{k_1} b_k - \left(Y_{k_1} - e_{k_1}\varepsilon_k\right)\right) \\ & + C_{k_2}\left(-e_{k_2}\right)^{\top} D_2\left(Y_{k_2} + e_{k_2}\varepsilon_k - \left(A_k w_k + e_{k_2} b_k\right)\right) \\ = & 0. \end{aligned} \tag{19}$$

Arranging the equations, we get:

$$\begin{bmatrix} w_k \\ b_k \end{bmatrix} = \left(L_k I + C_{k_1} G^{\top} D_1 G + C_{k_2} H D_2 H^{\top}\right)^{-1} \left(C_{k_1} G^{\top} D_1 f + C_{k_2} H^{\top} D_2 g\right). \tag{20}$$

Finally, the resulting equation is as (6). For the nonlinear form, we replace G and H above with (13). The whole training procedure of WLSMBSVR is summarized in Algorithm 2.

Algorithm 2 The training procedure for WLSMBSVR.

1: **Input**: The training sample set X, Y, parameters K, L_k, C_{k_1}, C_{k_2}, N, ε_k;
2: **Output**: The regression function $f(x)$;
3: Calculate $W_{i,j}$ and d_i for each point of samples in X using equation (14);
4: Divide X evenly into K parts, $X = [A_1; A_2; \cdots; A_k; \cdots; A_K]$, $Y = [Y_{1_1}; Y_{2_1}; \cdots; Y_{k_1}; \cdots; Y_{K_1}]$, **then** obtain the corresponding B_k, Y_{k_2} for each A_k, Y_{k_1};
5: Calculate D_1 and D_2 using equation (15);
6: **for** $k \le K$ **do**
  Obtain G and H from (13), according to (20), **then** get w_k and b_k;
  end for
7: According to (6), obtain the function $f(x)$.

3.3. The applications of WOA

GS is a standard method for parameter optimization. However, it is based on a discrete search. WOA can provide more refined parameter selection, potentially leading to better parameter combinations (Mirjalili and Lewis, 2016). On complex datasets, the advantages of WOA are more pronounced (Mostafa Bozorgi and Yazdani, 2019). Carbon stock estimates often include various environmental and biophysical factors that may have intricate interactions resulting in its complexity (Luo et al., 2020).

WOA solves optimization problems by simulating the social behavior and migration strategies of whales, including the search for optimal solutions or global minima (Mirjalili and Lewis, 2016). The theoretical part is as follows:

(i) The encirclement phase is described by:

$$\vec{X}(t+1) = \vec{X}^*(t) - \vec{A} \cdot \left|\vec{C} \cdot \left(\vec{X}^*(t) - \vec{X}(t)\right)\right|, \tag{21}$$

where t denotes the current iteration, $\vec{X}$ represents the position vectors of the whales. In the application of WOA to parameter optimization, $\vec{X}$ represents a set of parameters in the model. In WLSMBSVR, $\vec{X}$ is (K, L_k, C_k, C_{k_2}, N, ε_k, σ_k). X^* denotes the position vector of the optimal solution which is updated when a better solution is found, $\vec{A} = 2\vec{a} \cdot \vec{r} - \vec{a}$ and $\vec{C} = 2\vec{r}$ are the coefficient vectors, $\vec{a}$ is a vector that decreases linearly from 2 to 0 over the iterations, and $\vec{r}$ is a vector of random numbers in the range [0,1].

(ii) The spiral movement update is given by:

$$\vec{X}(t+1) = \vec{D}' \cdot e^{bl} \cdot \cos(2\pi l) + \vec{X}^*(t), \tag{22}$$

where $\vec{D}' = \left|\vec{X}^*(t) - \vec{X}(t)\right|$ defines the spatial separation between the hunting ground and the ith whale, b is a constant that shapes the logarithmic spiral trajectory, and l is a random number in the range [−1,1].

(iii) The combined behavior is modeled as:

$$\vec{X}(t+1) = \begin{cases} \vec{X}^*(t) - \vec{A} \cdot \vec{D}, \text{if } p < 0.5, \\ \vec{D}' \cdot e^{bl} \cos(2\pi l) + \vec{X}^*(t), \text{if } p \ge 0.5, \end{cases} \tag{23}$$

where p is a random number in the range [0,1]. Humpback whales search for prey based on their relative positions. Thus, the formula is adapted as follows:

$$\vec{D} = \left|\vec{C} \cdot \vec{X}_{\text{rand}} - \vec{X}\right| \quad \textit{and} \quad \vec{X}(t+1) = \vec{X}_{\text{rand}} - \vec{A} \cdot \vec{D}, \tag{24}$$

where $\vec{X}_{\text{rand}}$ is a randomly selected position from the current population.

Fig. 1 shows the specific operation flow chart of WOA combined with WLSMBSVR. Fig. 2 shows the evolution of WLSMBSVR. WOA iteratively refines the original model to find the optimal solution, and the time complexity and final performance primarily depend on the regression model used. Note that WLSMBSVR's runtime is higher than that of other models from the K-nearest neighbors graph. However,

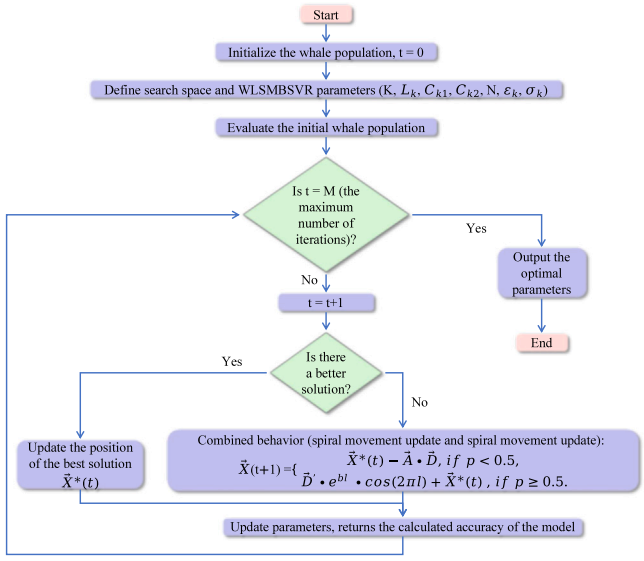


Fig. 1. The flowchart of WOA for the WLSMBSVR.

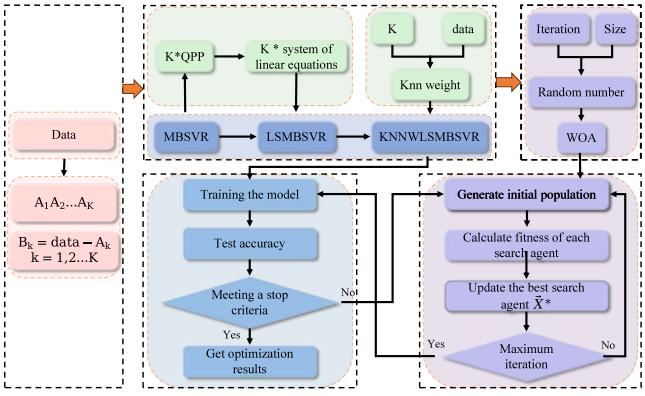


Fig. 2. Applying WOA in LSMBSVR and WLSMBSVR improves the efficiency and accuracy of parameter selection, further enhancing regression performance. The data is categorized into groups A and B, then fed into the WLSMBSVR for training. WOA is used to optimize the parameters of WLSMBSVR.

during the WLSMBSVR with WOA, only the inversion process is iteratively updated. So, when the number of iterations is sufficiently large, WLSMBSVR with WOA and LSMBSVR with WOA have similar time consumption levels, theoretically faster than SVR and TSVR, but with a significant enhancement in fitting performance.

4. Theoretical analysis and comparison

4.1. Comparative analysis of support vector regression models

The models above are based on the theoretical basis of SVM, that is, maximizing the interval to improve the generalization ability of the model. However, each model has its way of finding the optimal solution and its advantages or disadvantages. The following is an analysis of each model.

(1) SVR (Vapnik, 1995). SVR combines the ε-insensitive losses of upper and lower bounds in a single QPP, designed to minimize model complexity and control prediction errors. SVR is the most traditional support vector machine model, which has excellent regression performance and generalization ability. However, its computational efficiency is relatively low, especially with large data sets.

(2) TSVR (Peng, 2010). TSVR separates the ε-insensitive loss of the upper and lower bound into two small-scale QPPs, thus reducing the computation amount of each iteration, and the solution speed is theoretically four times faster than SVR.

(3) LSTSVR (Huang et al., 2013). LSTSVR uses the least squares method to solve two least squares problems instead of the traditional QPPs. This method uses the L_2 norm, transforming solving QPP problems into solving linear equations. The solution speed of LSTSVR is faster than SVR and TSVR but may result in reduced sparsity and increased sensitivity to outliers.

(4) MBSVR (Zhang et al., 2021). MBSVR divides the training set into K parts and constructs a hyperplane for each part by solving a QPP. The final regressor is the mean of the K hyperplanes. MBSVR is theoretically K^2 times faster than SVR.

(5) LSMBSVR. LSMBSVR solves the K group of linear equations by transforming inequality constraints into equality constraints using the least squares method. In terms of speed, linear equations are usually solved faster than QPPs. In terms of performance, LSMBSVR inherits the advantages of MBSVR, such as feasibility, superior fitting, and better regression performance.

(6) WLSMBSVR. WLSMBSVR uses KNN weight to solve K linear equations by minimizing the weighted squared error to get the optimal solution. WLSMBSVR enhances the ability to capture local structures to help deal with noise and outliers. The traditional least squares method may be more sensitive to outliers and increase the risk of overfitting, while WLSMBSVR can give the model better generalization performance by giving weights.

4.2. Computational complexity analysis

4.2.1. Computational complexity analysis of LSMBSVR

SVR, TSVR, and MBSVR primarily derive their solutions from solving QPPs. SVR addresses a larger QPP, while TSVR deals with two smaller QPPs. Theoretically, knowing from the comparison of their dual problems, the solution speed of TSVR is four times faster than SVR (Xu and Wang, 2014). For a certain QPP, MBSVR has the same number of constraints in its dual problem as TSVR, but it categorizes data into group A and group B, with only category A participating in the constraints. This makes MBSVR faster than TSVR for solving one QPP. Unlike SVR, TSVR, and MBSVR in solving QPPs, least squares methods obtain the optimal solution by solving linear equation systems, significantly enhancing solution speed. The time consumption of least squares models is primarily from matrix inversion. Since LSTSVR solves two linear systems, its time is roughly double that of LSSVR. More specifically, with the inversion complexity of LSSVR at $\mathcal{O}(l^3)$, and LSTSVR is at $\mathcal{O}(2(l+1)^3)$. The inversion complexity of LSMBSVR is at $\mathcal{O}(K(l+1)^3)$.

4.2.2. Computational complexity analysis of WLSMBSVR

WLSMBSVR involves two main computations:

(1) K-nearest neighbors graph calculation. Iterating over each point in the dataset and identifying its K closest neighbors from all points resulted in significant time consumption.

(2) Linear equation system calculation. WLSMBSVR and LSMBSVR have similar scales of matrix inversion, implying that their runtimes are approximately equivalent. This suggests that WLSMBSVR does not increase the computational complexity at linear equation system calculation.

5. Experiments evaluation of carbon stock datasets

5.1. Experimental setup

All experiments are conducted on a laptop with an i9-9880H CPU running at 2.3 GHz, 16 GB RAM, and a 512 GB hard drive using MATLAB R2021b.

5.1.1. Parameters

All six models employ the RBF kernel. For SVR, TSVR and LSTSVR, the parameters C (penalty term), σ (kernel term in RBF), and ϵ (breadth of the ϵ-tube) for all six models are set as $\{2^i \mid i = -9, -8, \ldots, 10\}$, $\{2^i \mid i = -9, -8, \ldots, 10\}$, and $\{i \times 0.1 \mid i = -9, -8, \ldots, 10\}$ through scientifically reasonable 5-fold cross-validation. MBSVR, LSMBSVR, and WLSMBSVR have additional parameters of the number of classes K and the regularization parameter L_k. The value of K is set to $\{i \mid i = 2, 3, \ldots, 10\}$. Each data point is randomly assigned an integer label from 1 to K as its classification label. L_k is set as $\{2^i \mid i = -9, -8, \ldots, 10\}$. The values of $C_{1_1}, C_{2_1}, \ldots, C_{K_1}$, $C_{1_2}, C_{2_2}, \ldots, C_{K_2}$, $\sigma_1, \sigma_2, \ldots, \sigma_K$, $\varepsilon_1, \varepsilon_2, \ldots, \varepsilon_K$, $L_1, L_2, \ldots, L_K$ and N are set to $C_{k_1} = \{2^i \mid i = -9, -8, \ldots, 10\}$, $C_{k_2} = \{2^i \mid i = -9, -8, \ldots, 10\}$, $\sigma_k = \{2^i \mid i = -9, -8, \ldots, 10\}$, $\varepsilon_k = \{i \times 0.1 \mid i = -9, -8, \ldots, 10\}$, $L_k = \{2^i \mid i = -9, -8, \ldots, 10\}$ and $N = 50$ through 5-fold cross-validation. The subsequent tables present the best R^2 and the corresponding indicators.

5.1.2. Evaluating indicators

Let l and n represent the number of data points in the training and testing sets, respectively. Denote y_i as the actual value for sample x_i, $\hat{y}_i$ as the corresponding predicted value, and $\bar{y}$ as the mean of the y_i values in the training set: $\bar{y} = \frac{1}{n}\sum_{i=1}^{n} y_i$. The upcoming metrics are utilized to assess the performance of the regression model.

- **SSE:** The total of squared differences between the actual and predicted values for the test data, calculated using the formula $SSE = \sum_{i=1}^{n}(y_i - \hat{y}_i)^2$.
- **SST:** The total variance of the test data, measured by the formula $SST = \sum_{i=1}^{n}(y_i - \bar{y})^2$.
- **SSR:** Sum squared deviation that can be explained by the estimator, which is defined as $SSR = \sum_{i=1}^{n}(\hat{y}_i - \bar{y})^2$.
- **SSR/SST:** A metric comparing the explained variance to the total variance in the test data, expressed as $SSR/SST = \sum_{i=1}^{n}\left(\hat{y}_i - \bar{y}\right)^2 / \sum_{i=1}^{n}\left(y_i - \bar{y}\right)^2$.
- **SSE/SST:** A measure of the unexplained variance relative to the total variance in the test data, defined as $SSE/SST = \sum_{i=1}^{n}\left(y_i - \hat{y}_i\right)^2 / \sum_{i=1}^{n}\left(y_i - \bar{y}\right)^2$.
- R^2: A statistical measure that represents the fraction of the variance for the dependent variable that is predictable from the independent variables in a regression analysis, calculated using $R^2 = 1 - SSE/SST = 1 - \sum_{i=1}^{n}\left(y_i - \hat{y}_i\right) / \sum_{i=1}^{n}\left(y_i - \bar{y}\right)^2$.
- **MAE:** MAE finds the average of the absolute differences between actual and predicted data points, calculated using MAE = $\frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|$.
- **RMSE:** RMSE takes the square root of the average of the squared differences, calculated using RMSE = $\sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}$.

A lower SSE/SST ratio typically suggests a more accurate model fit, while a higher SSR/SST ratio indicates a greater proportion of variance explained. An SSE/SST value that approaches zero may signal overfitting. An effective regression model should balance these two metrics. Our experiment uses these two indicators to measure simultaneously.

5.2. Application of GS on carbon storage datasets

From Table 3, it can be observed that the LSMBSVR has a significant advantage in computational time while also presenting relatively good regression performance. The WLSMBSVR does not show a notable advantage in computational time, but it exhibits the best predictive performance in regression. WLSMBSVR achieves superior performance in the ORCd and ORD datasets, particularly excelling in ORCd.

In Fig. 3, green dots are the KNN-calculated weights and corresponding observed values of the test samples, while red dots are the KNN-calculated weights and corresponding predicted values. WLSMBSVR demonstrates stronger predictive robustness at higher-weight points and better generalization performance at lower-weight points.

Fig. 4 shows the detailed performance on carbon stock datasets using GS. Each blue dot represents the actual observed value of the test sample with the predicted value of the corresponding method. The black lines are point-to-point lines where the predicted values are the same as the actual values. The red lines represent the line fitting of these blue dots. Compared to other models, the fitting line of LSMBSVR and WLSMBSVR coincide more with the black lines. This suggests that the predictive trends are highly consistent with the observed trends. The points are tightly clustered around the decision boundary, which signifies that LSMBSVR and WLSMBSVR have minor predictive errors, implying a low predictive bias.

5.3. Application of WOA on carbon storage datasets

Let M denote the maximum iteration count, S the initial population size, and X_{lb}, X_{ub} represent vectors of all parameters' lower and upper bounds. Where M and S are set as $\{10 \times i \mid i = 1, 2, \ldots, 5\}$ and $\{5 \times i \mid i = 1, 2, \ldots, 10\}$. Specifically, the settings for X_{lb} and X_{ub} are determined as follows: for each regressor, we document the optimal parameters obtained from each GS, get the upper and lower ranges, and then define the actual optimization ranges to meet the following conditions: (1) The boundaries are set beyond the range of all optimal parameters found in the GS. (2) They are the closest integer powers of 2 to the lower and upper limits, where X_{lb} and X_{ub} are powers of 2. The WOA substantially reduces the time required for parameter search while optimizing the parameters. The subsequent tables present the best R^2 and the corresponding indicators through 5-fold cross-validation.

From Table 4, the WOA leads to performance improvements across all models. Notely, WLSMBSVR initially performs poorly on the ORDH dataset but achieves the best outcomes after optimization of WOA.

Fig. 5 shows the detail performance using WOA. Fig. 6 demonstrates that the WLSMBSVR exhibits the most excellent generalization ability on the ORCd dataset after WOA. Fig. 7 shows that R^2 increases are observed for all models on the carbon storage datasets after WOA. LSMBSVR and WLSMBSVR experience significant optimization, showing a higher enhancement.

6. Experiment validation

To verify that our models have application value in more scenarios, we extensively test SVR, TSVR, LSTSVR, MBSVR, LSMBSVR, and WLSMBSVR across four synthetic datasets, three synthetic datasets with appropriate interference points, and seven UCI datasets. The experimental setup is consistent with Section 5.

6.1. Experiments on synthetic data

6.1.1. Synthetic datasets

We utilize the sinc function to compare the performance, which is defined as

$$f(x) = \frac{\sin x}{x}.$$

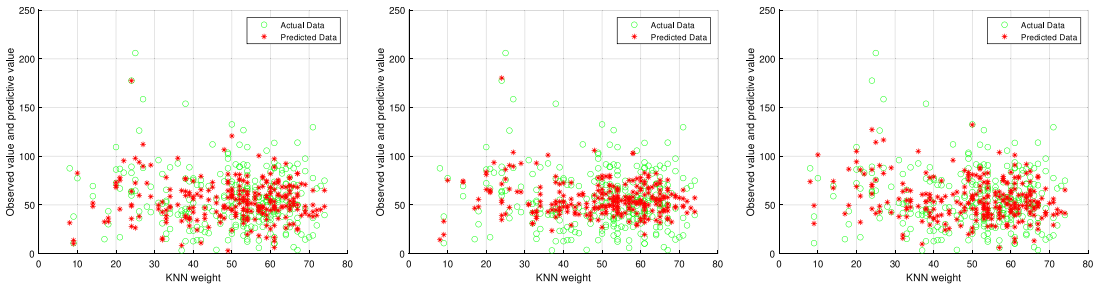
Introducing Gaussian noise of varying variance, each point with the corresponding parameters as:

$$\text{(TypeA)} \quad y_i = f(x_i) + \eta_i, \quad x_i \sim U[-12, 12], \quad \eta_i \sim N\left(0, 0.1^2\right),$$

$$\text{(TypeB)} \quad y_i = f(x_i) + \eta_i, \quad x_i \sim U[-12, 12], \quad \eta_i \sim N\left(0, 0.2^2\right),$$

where η_i represents the specified Gaussian noise, $U[a, b]$ stands for a uniformly distributed random variable within the range $[a, b]$, and $N(c, d^2)$ denotes a Gaussian random variable with mean c and variance d^2. We use the following as another objective function:

$$g(x) = \frac{|x-1|}{4} + \left|\sin\left(\pi + \frac{\pi(x-1)}{4}\right)\right| + 1.$$



(a) SVR. (b) MBSVR. (c) WLSMBSVR.

Fig. 3. K-nearest neighbors and corresponding actual data or predicted data of test points for SVR, MBSVR, and WLSMBSVR on ORCd using GS.

Table 3
Regression performance comparison of SVR, TSVR, LSTSVR, MBSVR, LSMBSVR, and WLSMBSVR on carbon stock dataset using GS.

Database	Regressor	MAE	RMSE	SSE/SST	SSR/SST	R^2	Time
ORCd	SVR	18.1058	24.9932	0.69255	0.4833	0.3074	0.1266
	TSVR	17.0484	22.0566	0.53935	0.5	0.4606	0.5062
	LSTSVR	16.7783	22.361	0.55436	0.3605	0.4456	0.1067
	MBSVR	20.1821	26.6012	0.78454	0.3308	0.2154	0.5841
	LSMBSVR	17.7455	23.4197	0.6081	0.2385	0.3919	0.1328
	WLSMBSVR	16.5262	22.0517	0.53913	0.4594	**0.4609**	0.4417
ORDH	SVR	1.7445	2.4971	0.29243	0.8674	**0.7076**	0.1387
	TSVR	1.7107	2.5158	0.29682	0.7718	0.7032	0.7256
	LSTSVR	1.7245	2.5339	0.30111	0.7649	0.6989	0.1123
	MBSVR	1.7375	2.5301	0.30021	0.7163	0.6998	0.5967
	LSMBSVR	1.7123	2.5184	0.29742	0.7095	0.7026	0.0998
	WLSMBSVR	1.7392	2.554	0.30589	0.7043	0.6941	0.3364
ORD	SVR	2.2671	2.9636	0.44673	0.6711	0.5533	0.1371
	TSVR	2.256	2.9359	0.43843	0.7014	0.5616	0.5271
	LSTSVR	2.2565	2.9446	0.44103	0.6954	0.5589	0.1099
	MBSVR	2.2256	2.9304	0.43679	0.5788	0.5632	0.5807
	LSMBSVR	2.2379	2.9158	0.43244	0.6115	0.5676	0.1261
	WLSMBSVR	2.231	2.9054	0.42938	0.5667	**0.5706**	0.3423

Table 4
Regression performance comparison of SVR, TSVR, LSTSVR, MBSVR, LSMBSVR, and WLSMBSVR on carbon stock dataset using WOA.

Database	Regressor	MAE	RMSE	SSE/SST	SSR/SST	R^2
ORCd	SVR	18.5472	24.6149	0.6752	0.1789	0.3254
	TSVR	16.3149	21.54	0.4965	0.3041	0.5035
	LSTSVR	16.6736	21.4811	0.5471	0.268	0.4529
	MBSVR	18.5374	23.6538	0.7439	0.1355	0.2561
	LSMBSVR	15.9388	21.9365	0.5791	0.3368	0.4209
	WLSMBSVR	16.0503	21.3854	0.4729	0.395	**0.5271**
ORDH	SVR	1.7298	2.5918	0.2689	0.709	0.731
	TSVR	1.6177	2.7413	0.2784	0.6203	0.7215
	LSTSVR	1.9113	2.9164	0.2911	0.6814	0.7089
	MBSVR	1.8685	2.6844	0.2835	0.7311	0.7165
	LSMBSVR	1.76	2.6091	0.2763	0.7533	0.7237
	WLSMBSVR	1.728	2.695	0.2556	0.6672	**0.7444**
ORD	SVR	1.5228	6.261	0.4155	0.4059	0.5845
	TSVR	1.5396	2.5129	0.3823	0.6067	0.6177
	LSTSVR	1.7817	2.7819	0.4367	0.6101	0.5633
	MBSVR	1.6056	2.6292	0.4131	0.6186	0.5869
	LSMBSVR	1.7093	2.8235	0.3679	0.6942	0.6321
	WLSMBSVR	1.465	2.3254	0.3348	0.6832	**0.6652**

We add the following different forms of Gaussian noise:

(TypeC) $y_i = g(x_i) + \eta_i, \quad x_i \sim U[-10, 10], \quad \eta_i \sim N\left(0, 0.2^2\right),$

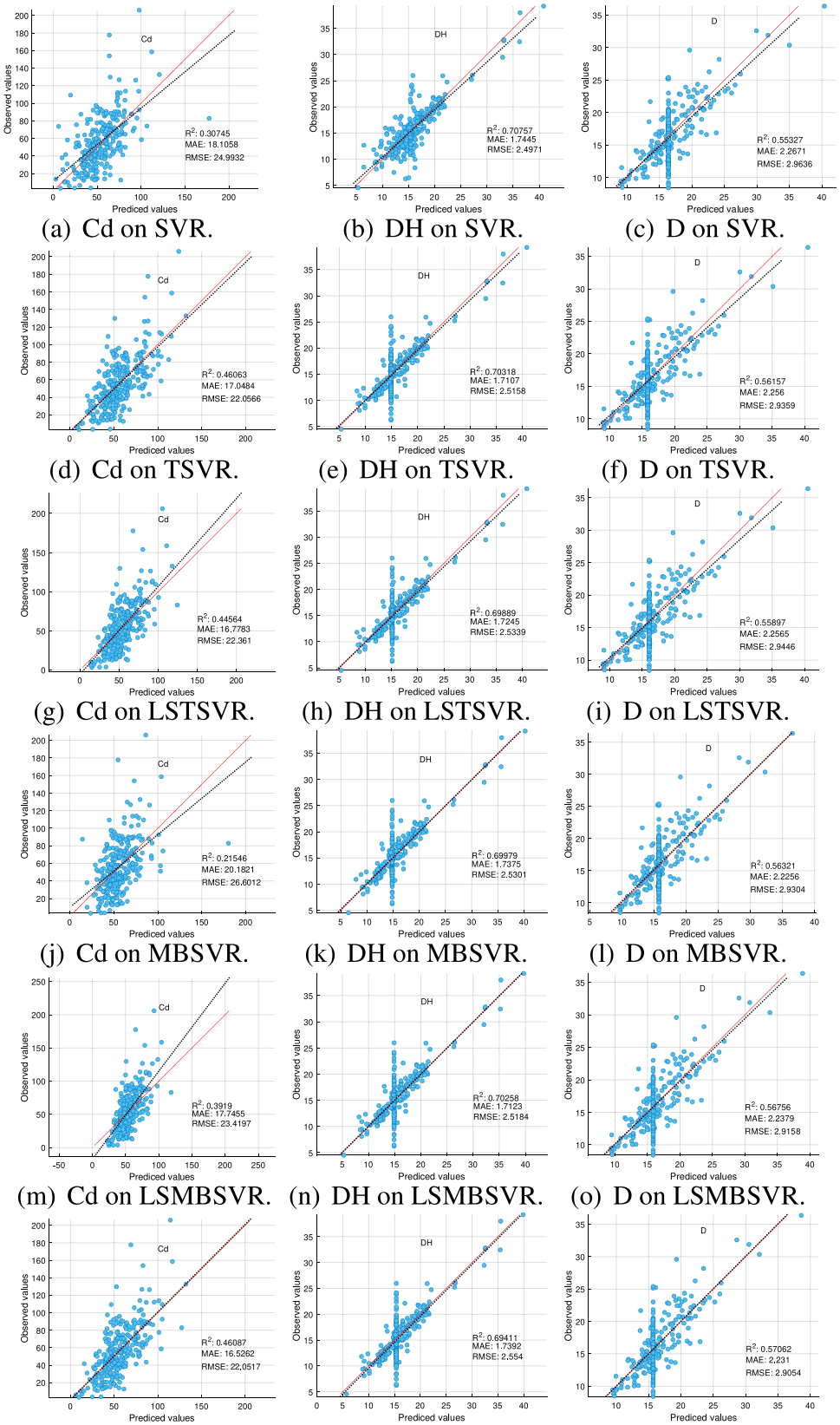
(TypeD) $y_i = g(x_i) + \eta_i, \quad x_i \sim U[-10, 10], \quad \eta_i \sim N\left(0, 0.4^2\right).$

For noise TypeA and TypeB, the training set comprises 1000 samples, and the testing set comprises 500 samples. The training and testing sets each contain 1000 samples for noise TypeC and TypeD.

6.1.2. Results on synthetic datasets

In Table 5, in terms of SSE and SSE/SST, among types A, B, and C, WLSMBSVR and LSMBSVR outperform compared models, where WLSMBSVR shows a slight advantage over LSMBSVR. Only TypeD is inferior to SVR. Regarding SSR/SST, WLSMBSVR and LSMBSVR are similar to the compared models. In terms of the time, although LSMBSVR is slower than LSTSVR, it offers considerable performance improvements and is generally faster than the remaining models. WLSMBSVR is generally slower than compared models due to the high computational cost of the KNN. However, WLSMBSVR may provide best performance.

We can visually demonstrate the fitting of one-dimensional feature data by using two-dimensional images of the model curve, where the red dots represent the coordinates of each sample point (x_i, y_i). Each curve in the figure shows the fitting effect of different models in the corresponding interval. Figs. 8(a) and 8(b) elucidate that LSMBSVR



(a) Cd on SVR. (b) DH on SVR. (c) D on SVR.

(d) Cd on TSVR. (e) DH on TSVR. (f) D on TSVR.

(g) Cd on LSTSVR. (h) DH on LSTSVR. (i) D on LSTSVR.

(j) Cd on MBSVR. (k) DH on MBSVR. (l) D on MBSVR.

(m) Cd on LSMBSVR. (n) DH on LSMBSVR. (o) D on LSMBSVR.

(p) Cd on WLSMB-SVR. (q) DH on WLSMB-SVR. (r) D on WLSMBSVR.

Fig. 4. Comparative analysis of SVR, TSVR, LSTSVR, MBSVR, LSMBSVR, and WLSMBSVR models on carbon stock datasets using GS.

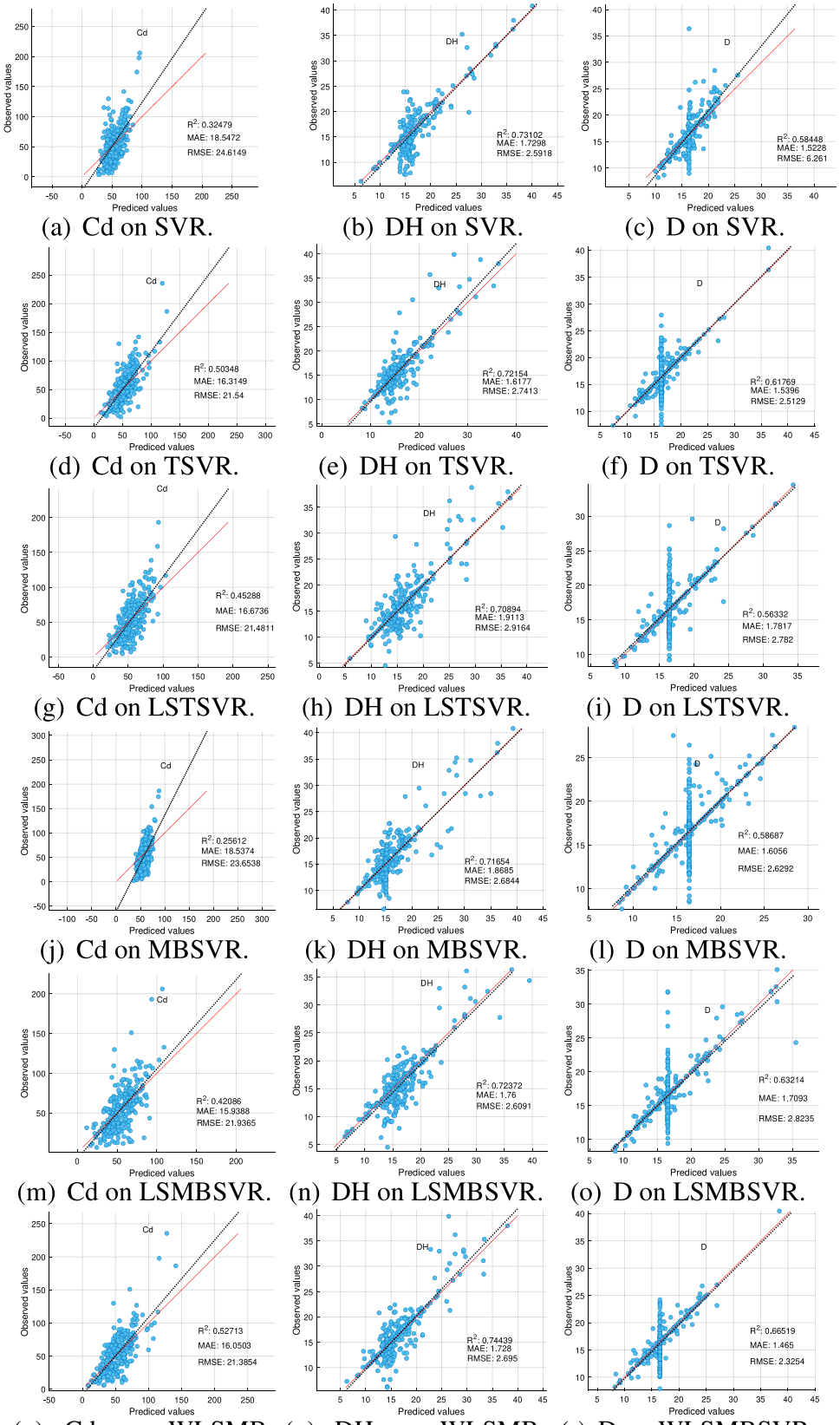
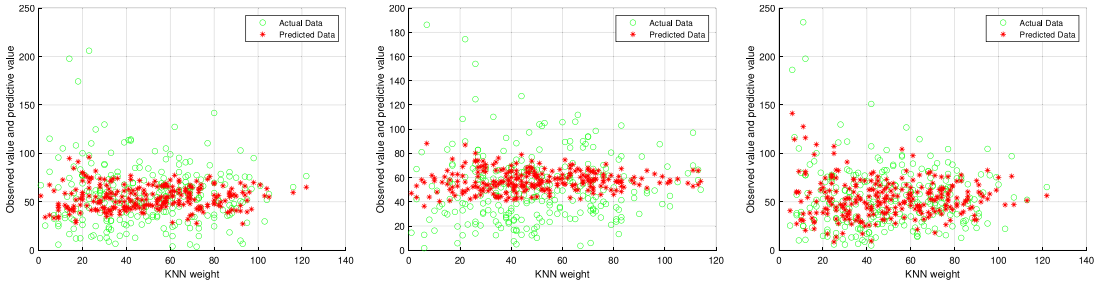


Fig. 5. Comparative analysis of SVR, TSVR, LSTSVR, MBSVR, LSMBSVR, and WLSMBSVR models on carbon stock datasets using WOA.

Table 5
Result comparison of SVR, TSVR, LSTSVR, MBSVR, LSMBSVR and WLSMBSVR on four synthetic datasets with different forms of noise.

Noise	Regressor	SSE	SSE/SST	SSR/SST	Time
TypeA	SVR	3.8048 ± 0.0514	0.0902 ± 0.0082	0.9116 ± 0.0020	1.2030 ± 0.0021
	TSVR	3.7680 ± 0.0835	0.0892 ± 0.0008	0.9269 ± 0.0045	0.2677 ± 0.0023
	LSTSVR	3.7848 ± 0.0785	0.0897 ± 0.0008	0.9173 ± 0.0021	0.0583 ± 0.0048
	MBSVR	3.7508 ± 0.0672	0.0888 ± 0.0007	0.9212 ± 0.0025	0.2130 ± 0.0097
	LSMBSVR	3.7523 ± 0.0671	0.0888 ± 0.0007	0.9131 ± 0.0026	0.0779 ± 0.0016
	WLSMBSVR	3.7522 ± 0.0669	0.0888 ± 0.0007	0.9131 ± 0.0027	0.70923 ± 0.0033
TypeB	SVR	16.1069 ± 0.86205	0.2976 ± 0.0078	0.6857 ± 0.0018	0.81038 ± 0.0004
	TSVR	15.0783 ± 1.1551	0.2782 ± 0.0061	0.7451 ± 0.0085	0.2109 ± 0.0014
	LSTSVR	15.1391 ± 1.2562	0.2794 ± 0.0067	0.7395 ± 0.0015	0.0458 ± 0.0000
	MBSVR	15.1794 ± 1.0410	0.2798 ± 0.0045	0.7450 ± 0.0018	0.2289 ± 0.0000
	LSMBSVR	15.0136 ± 1.0704	0.2770 ± 0.0058	0.7345 ± 0.0019	0.10352 ± 0.0001
	WLSMBSVR	15.0135 ± 1.0845	0.2770 ± 0.0057	0.7366 ± 0.0028	0.73068 ± 0.0000
TypeC	SVR	45.802 ± 3.9701	0.1489 ± 0.0044	0.8481 ± 0.0015	0.9270 ± 0.0086
	TSVR	13.859 ± 0.7820	0.0451 ± 0.0009	0.9649 ± 0.0031	0.1936 ± 0.0035
	LSTSVR	13.891 ± 0.7539	0.0452 ± 0.0009	0.9648 ± 0.0030	0.0461 ± 0.0007
	MBSVR	14.000 ± 0.4688	0.0455 ± 0.0005	0.9625 ± 0.0024	0.2359 ± 0.0010
	LSMBSVR	14.032 ± 0.5844	0.0456 ± 0.0007	0.9629 ± 0.0022	0.1193 ± 0.0026
	WLSMBSVR	14.032 ± 0.5849	0.0456 ± 0.0008	0.9629 ± 0.0022	0.9055 ± 0.0011
TypeD	**SVR**	63.9100 ± 4.4727	0.1449 ± 0.0022	0.8480 ± 0.0011	0.7901 ± 0.0030
	TSVR	55.0550 ± 13.1091	0.1608 ± 0.0015	0.8711 ± 0.0089	0.2048 ± 0.0006
	LSTSVR	55.2251 ± 12.6299	0.1613 ± 0.0014	0.8715 ± 0.0086	0.0480 ± 0.0045
	MBSVR	55.4958 ± 8.5149	0.1622 ± 0.0013	0.8690 ± 0.0069	0.3787 ± 0.0024
	LSMBSVR	55.185 ± 7.6142	0.1603 ± 0.0012	0.8623 ± 0.0089	0.0788 ± 0.0013
	WLSMBSVR	55.167 ± 7.6143	0.1603 ± 0.0012	0.8623 ± 0.0089	0.7757 ± 0.0016



(a) SVR. (b) MBSVR. (c) WLSMBSVR.

Fig. 6. K-nearest neighbors and corresponding actual data or predicted data of test points for SVR, MBSVR, and WLSMBSVR on ORCd after WOA.

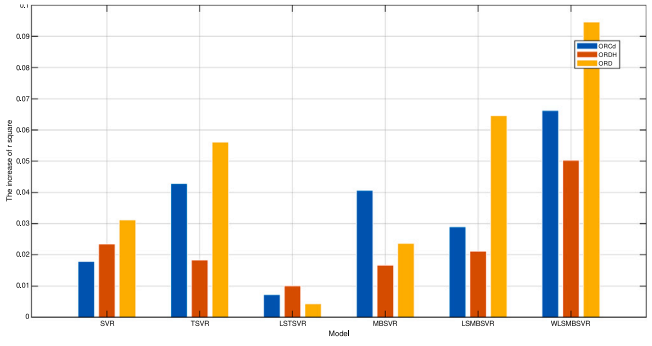


Fig. 7. The growth of R^2 of carbon storage data after WOA.

and MBSVR exhibit comparable performance in capturing patterns across all points corrupted by diverse Gaussian noise. Particularly, WLSMBSVR shows superior fitting performance.

Figs. 8(c) and 8(d) vividly demonstrate the robustness of LSMBSVR and WLSMBSVR, particularly the latter, in regions with sharp multiple inflections. LSMBSVR and WLSMBSVR generally yield favorable fits on artificial datasets and execute efficiently. It can be shown that WLSMBSVR excels in precise predictive accuracy and resilience against noise.

6.2. Experiments on UCI data

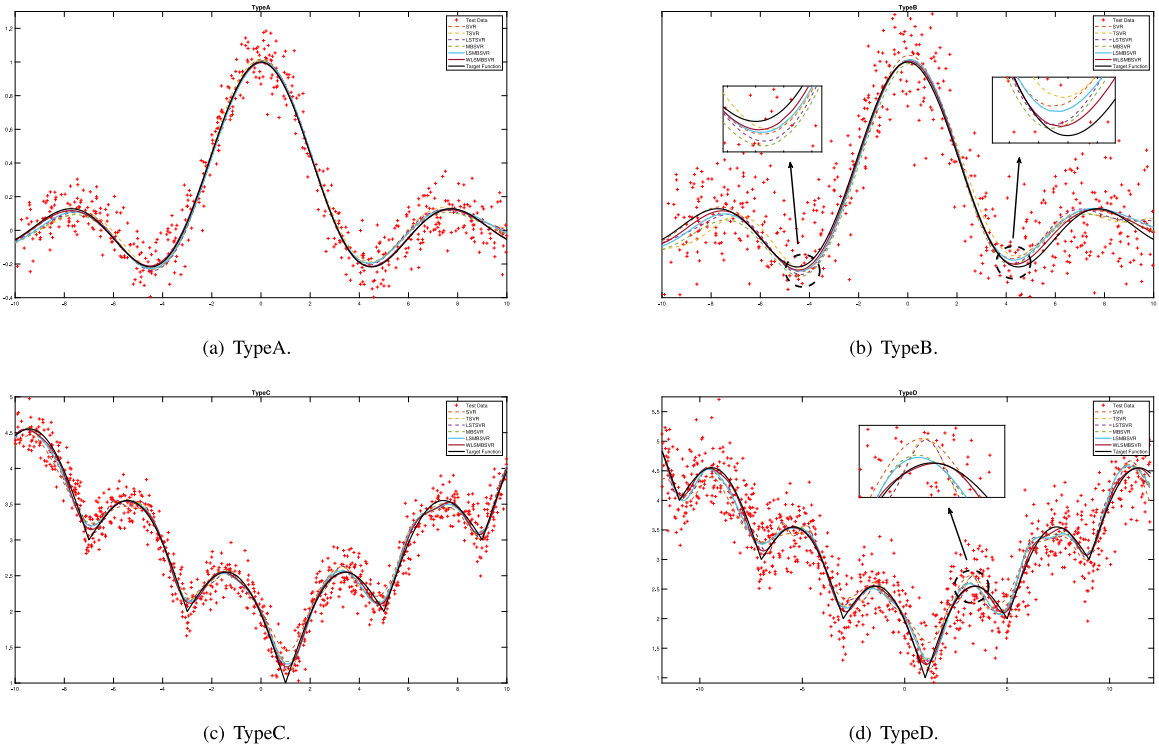
LSMBSVR and WLSMBSVR exhibit distinct advantages presented in Table 6. LSMBSVR provides robust estimate accuracy while showing relatively faster computational times across various datasets. Meanwhile, WLSMBSVR stands out for its exceptional accuracy, consistently outperforming compared models in datasets.

6.3. The application of WOA

During this experiment, we use TypeC, LiveDiosders, Parkinson and AtuoMpg as representatives for testing. From Table 7, it can be observed that after applying WOA, WLSMBSVR demonstrates the best performance among all models. Additionally, in Fig. 9, LSMBSVR and WLSMBSVR still show a significant improvement in optimization after applying WOA compared to other models.

6.4. Parameter analysis

Figs. 10 and 11 show the parameter sensitivity analysis of LSMBSVR and WLSMBSVR. Figs. 10 indicates that the parameters of K and L_k in LSMBSVR have a significant impact on the final R^2, therefore, careful adjustment of these parameters is necessary. The same situation goes for K and C_{k_2} in WLSMBSVR from Figs. 11. Adding C_{k_2} and K parameters in WLSMBSVR increases the complexity and enhances its tunability. This provides a better opportunity for the WOA.



(a) TypeA. (b) TypeB.

(c) TypeC. (d) TypeD.

Fig. 8. Fitting curves of six mentioned models on noise datasets.

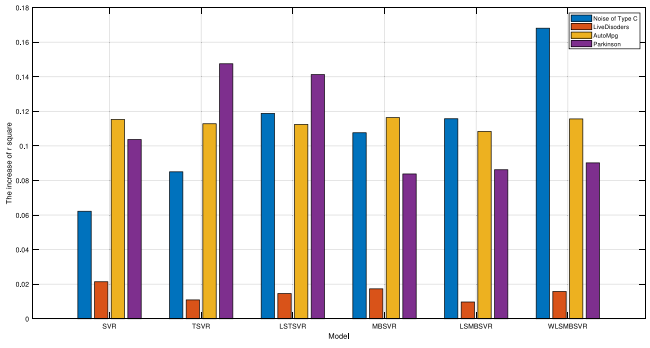


Fig. 9. The growth of R^2 of diverse datasets after WOA.

7. Discussion and analysis

7.1. Discussion on generalization ability

Occam's razor is a central tenet of scientific research that states that when multiple hypotheses can explain an observation, the simplest one should be chosen. We equate "smoother" with "simpler". The graphical analysis shows that WLSMBSVR generally behaves smoother and simpler than other models. This is because WLSMBSVR can capture local information of the sample and give more weight to important points. WLSMBSVR is smoother and can better learn the general rules of carbon storage estimation tasks. As a result, WLSMBSVR is more effective at defending against interference points and is more likely to perform better on new samples than those "scraggly" models, demonstrating greater generalization.

From Figs. 3 and 6 known, WLSMBSVR demonstrates stronger predictive robustness at points with higher weights. Moreover, there is an increased likelihood of noise and interference at points with fewer weights. By giving them small weights, WLSMBSVR exhibits better generalization capabilities. The KNN algorithm focuses on the local information of samples, which helps capture local data patterns. Therefore, WLSMBSVR performed excellently on ORCd. However, the MBSVR shows a more overfitting to specific samples.

From the estimates of DH in Figs. 4 and 5, it can be shown that many observed values are predicted to be the same, while after optimization of WOA, this issue is mitigated, indicating the parameters of the model are better adjusted. The case of many observed values being predicted to be the same for D is more severe, which is still severe after WOA. This may be due to the models being overfitted. These models perform well on the training set, but the prediction results are simple on the test set, which also indicates that these models have poor generalization ability. In this case, LSMBSVR and WLSMBSVR have less overfitting and superior generalization ability than other models.

7.2. Discussion on anti-interference capability

The anti-interference ability of the model is the ability of the model to maintain its performance in the face of noise, outliers, incomplete information, or hostile attacks in the input data. A model with strong generalization ability can learn the intrinsic laws of the data from the training data, rather than just remembering the training sample. Our model gives more weight to more important samples and can better grasp important sample information, so our model has the anti-interference ability.

From Fig. 8 known, at the inflection point, regions susceptible to overfitting due to limited data availability. But LSMBSVR and, notably, WLSMBSVR exhibit a high degree of fitting without overfitting, especially in areas with higher noise levels, highlighting their robustness.

We experiment with synthetic datasets by adding outliers to assess the anti-interference capability. We add three "Outliers" to TypeA, five "Outliers" and one "Outliers" to TypeC. Figs. 12, 13 and 14 show that TSVR experiences notable fluctuations in the presence of the "Outliers", indicating a sensitivity to the outlier. In comparison, LSMBSVR and WLSMBSVR exhibit less fluctuation, suggesting a more stable performance. Particularly, WLSMBSVR shows almost a consistent

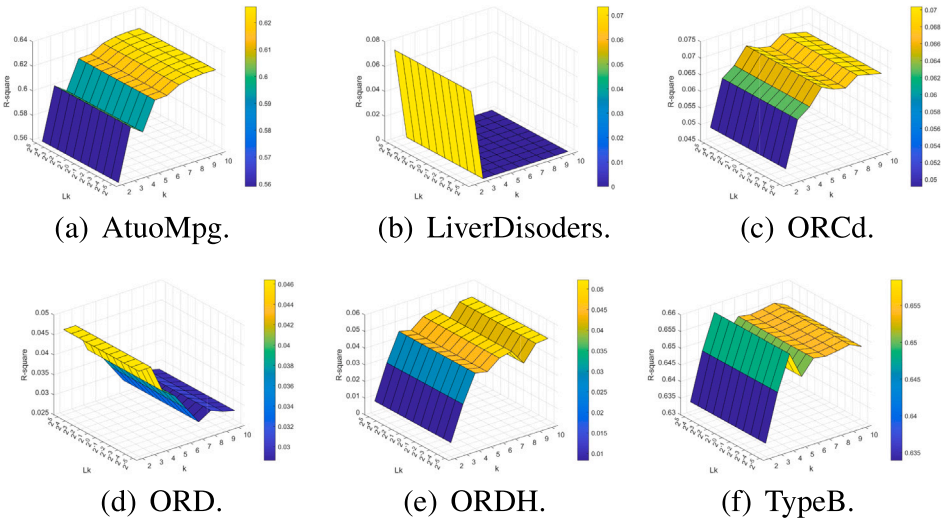


Fig. 10. Sensitivity testing of L_k and K in LSMBSVR on various datasets.

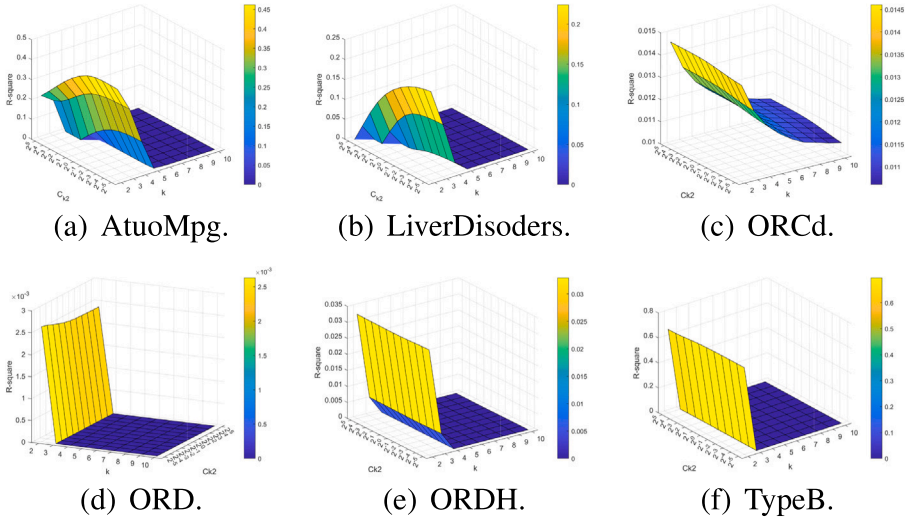


Fig. 11. Sensitivity testing of C_{k_2} and K in WLSMBSVR on various datasets.

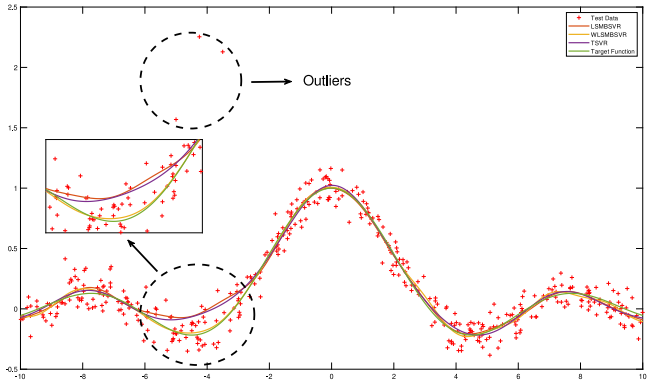


Fig. 12. Performance comparison of TSVR, LSMBSVR, and WLSMBSVR on noise TypeA with three "Outliers".

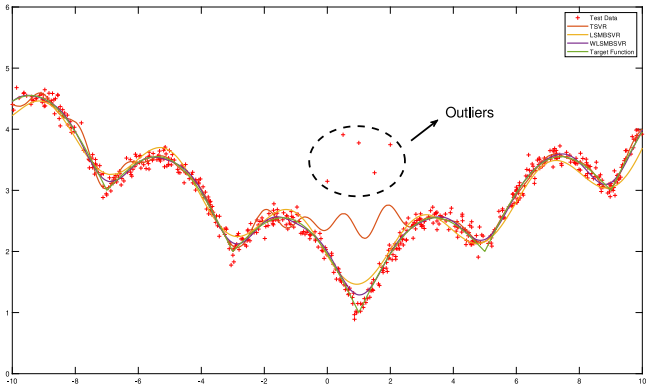


Fig. 13. Performance comparison of TSVR, LSMBSVR, and WLSMBSVR on noise TypeC with five "Outliers".

Table 6
Result comparison of six models on UCI datasets.

Database	Regressor	SSE/SST	SSR/SST	Time
Parkinson	SVR	0.0836 ± 0.0040	0.8667 ± 0.0024	1.9325 ± 0.0069
	TSVR	0.1023 ± 0.0041	0.9608 ± 0.0067	0.1938 ± 0.0078
	LSTSVR	0.1135 ± 0.0022	0.9862 ± 0.0017	0.0370 ± 0.0020
	MBSVR	0.0844 ± 0.0018	0.8516 ± 0.0031	0.2239 ± 0.0027
	LSMBSVR	0.0756 ± 0.0085	0.8549 ± 0.0058	0.0984 ± 0.0012
	WLSMBSVR	0.0708 ± 0.0001	0.8672 ± 0.0044	0.8235 ± 0.0072
AutoMpg	SVR	0.3974 ± 0.0015	0.9731 ± 0.0012	6.0380 ± 0.0035
	TSVR	0.3182 ± 0.0037	0.7967 ± 0.0013	0.0472 ± 0.0054
	LSTSVR	0.3189 ± 0.0034	0.7980 ± 0.0014	0.0101 ± 0.0016
	MBSVR	0.3052 ± 0.0034	0.6862 ± 0.0025	0.0502 ± 0.0015
	LSMBSVR	0.3041 ± 0.0058	0.7147 ± 0.0086	0.0129 ± 0.0034
	WLSMBSVR	0.3035 ± 0.0012	0.7398 ± 0.0068	0.1287 ± 0.0090
AirQuality	SVR	0.0427 ± 0.0040	0.9526 ± 0.0025	241.9018 ± 0.5605
	TSVR	0.0218 ± 0.0024	0.9799 ± 0.0011	88.4634 ± 7.5036
	LSTSVR	0.0367 ± 0.0026	0.9643 ± 0.0080	21.7283 ± 0.1373
	MBSVR	0.0394 ± 0.0050	0.9526 ± 0.0026	60.3627 ± 2.2731
	LSMBSVR	0.0373 ± 0.0040	0.9601 ± 0.0037	27.7927 ± 0.2479
	WLSMBSVR	0.0202 ± 0.0022	0.9960 ± 0.0020	233.2978 ± 81.7306
BikeSharing	SVR	0.1210 ± 0.0010	0.8565 ± 0.0033	0.4132 ± 0.0000
	TSVR	0.1249 ± 0.0018	0.9038 ± 0.0057	0.5070 ± 0.0000
	LSTSVR	0.1268 ± 0.0039	0.9502 ± 0.0037	0.0255 ± 0.0000
	MBSVR	0.1153 ± 0.0013	0.8677 ± 0.0077	0.1022 ± 0.0002
	LSMBSVR	0.1150 ± 0.0013	0.8740 ± 0.0086	0.0524 ± 0.0000
	WLSMBSVR	0.1144 ± 0.0013	0.8742 ± 0.0093	0.4435 ± 0.0003
Forestfires	SVR	0.1597 ± 0.0029	0.8795 ± 0.0098	0.1642 ± 0.0000
	TSVR	0.1477 ± 0.0054	0.8541 ± 0.0017	0.0759 ± 0.0000
	LSTSVR	0.1413 ± 0.0014	0.8526 ± 0.0017	0.0164 ± 0.0000
	MBSVR	0.0943 ± 0.0070	0.8874 ± 0.0060	0.0732 ± 0.0000
	LSMBSVR	0.0862 ± 0.0065	0.8877 ± 0.0043	0.0263 ± 0.0000
	WLSMBSVR	0.0819 ± 0.0035	0.9254 ± 0.0026	0.1590 ± 0.0008
LiverDisoders	SVR	0.7163 ± 0.0037	0.2794 ± 0.0058	0.1070 ± 0.0000
	TSVR	0.7040 ± 0.0074	0.4085 ± 0.0130	0.0256 ± 0.0000
	LSTSVR	0.7379 ± 0.0081	0.4922 ± 0.0198	0.0053 ± 0.0000
	MBSVR	0.6600 ± 0.0034	0.3394 ± 0.0146	0.0240 ± 0.0000
	LSMBSVR	0.6600 ± 0.0035	0.3647 ± 0.0173	0.0096 ± 0.0000
	WLSMBSVR	0.6590 ± 0.0037	0.3655 ± 0.0172	0.1063 ± 0.0000
SeoulBikeData	SVR	0.3549 ± 0.0044	0.7402 ± 0.0050	122.1166 ± 0.0195
	TSVR	0.3964 ± 0.0012	0.6013 ± 0.0047	74.6178 ± 6.8376
	LSTSVR	0.3994 ± 0.0010	0.5932 ± 0.0047	17.8467 ± 0.0069
	MBSVR	0.2989 ± 0.0030	0.7104 ± 0.0029	52.7588 ± 1.7942
	LSMBSVR	0.2942 ± 0.0035	0.7453 ± 0.0024	22.5014 ± 1.1315
	WLSMBSVR	0.2701 ± 0.0035	0.7187 ± 0.0075	148.9756 ± 20.8291

Table 7
R^2 comparison of six models with GS and WOA.

Regressor	Method	TypeC	LiveDisoders	Parkinson	AtuoMpg
SVR	GS	0.2837	0.7024	0.8424	0.6880
	WOA	0.3459	0.7238	0.9461	0.8032
TSVR	GS	0.2960	0.7218	0.8525	0.6989
	WOA	0.3810	0.7327	1.0000	0.8117
LSTSVR	GS	0.2622	0.7206	0.8587	0.6965
	WOA	0.3573	0.7352	1.0000	0.8089
MBSVR	GS	0.3400	0.7202	0.9057	0.6930
	WOA	0.4476	0.7375	0.9894	0.8093
LSMBSVR	GS	0.3400	0.7230	0.9138	0.7008
	WOA	0.4557	0.7327	1.0000	0.8092
WLSMBSVR	GS	0.3409	0.7230	0.9140	0.7010
	WOA	0.5090	0.7388	1.0000	0.8119

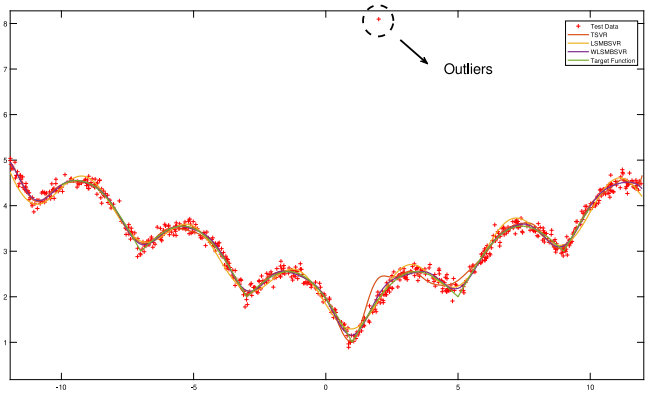


Fig. 14. Performance comparison of TSVR, LSMBSVR, and WLSMBSVR on noise TypeC with single "Outliers".

fit to the target function, indicating a best robust resistance to the introduced noise. In general, least squares models are typically not sparse and are sensitive to noise and outliers. However, The figure shows our models exhibit better anti-interference ability compared to other models.

Carbon stock estimation can be viewed as a regression problem, and therefore, our models are well suited for solving such problems. The carbon storage dataset comes from a wide range of sources, and the collection standards are not uniform, so interference and noise are inevitable. Our models take complete account of structural information and thus provide more robustness at points with higher weights. Outliers usually have fewer neighbors and are therefore assigned lower weights, which makes the models more resistant to noise

and interference. In addition, using WOA can further improve the effect of parameter optimization so that the models can find the optimal parameters faster, thereby improving the performance.

8. Conclusion and future works

For the carbon storage estimation aspect, we analyze the potential issues encountered in carbon storage estimation. We use the SVR framework for high-dimensional and nonlinear processing, KNN weighting to solve noise and interference, enhance robustness and generalization, and WOA for parameter selection. For the model aspect, we observe that MBSVR encounters challenges when dealing with noise and outliers. So, we propose LSMBSVR, which significantly improves regression speed while maintaining comparable regression performance, and WLSMBSVR, which enhances the resistance to interference. Also, we integrate the WOA for parameter optimization, further improving the fitting performance. The improved models in this study achieve satisfactory results and provide new perspectives and methods for solving regression problems. Many models have performed well in imaging data to monitor biomass (Liu et al., 2024b; Liu et al., 2024a). Our model holds potential in image data monitoring, particularly in remote sensing and satellite imagery. These technologies provide data on biomass and vegetation cover, and our model offers enhanced precision and interference resistance in predicting carbon reserves based on these data. Furthermore, the model can be integrated into computational platforms for estimating carbon and biomass. Also, we can further refine the model to achieve time-series analysis and improve predictive accuracy. Such integration and enhancement will provide robust technical support for environmental monitoring and carbon management.

While our models provide excellent resistance to interference and are more robust and generalized, they still face overfitting problems when applied to practical, particularly complex datasets, leading to poor generalization. Thus, it is essential to improve the generalization capability in real-world applications. Our future work will focus on integrating our models with ensemble learning, optimizing feature engineering, adapting to complex data structures, handling large-scale data, exploring the potential applications of deep learning, and validating model performance in real-world scenarios. This comprehensive approach aims to further enhance the capability of solving regression problems, promoting universality and practicality and continuously advancing the field of regression problems.

CRediT authorship contribution statement

Niannian Deng: Writing – original draft, Visualization, Validation, Software, Investigation, Formal analysis, Data curation, Conceptualization. **Renpeng Xu:** Writing – original draft, Visualization, Validation, Software, Investigation, Formal analysis, Data curation, Conceptualization. **Ying Zhang:** Writing – original draft, Visualization, Validation, Software, Formal analysis, Data curation, Conceptualization. **Haoting Wang:** Writing – original draft, Visualization, Validation, Software, Formal analysis, Data curation, Conceptualization. **Chen Chen:** Writing – original draft, Formal analysis, Conceptualization. **Huiru Wang:** Writing – review & editing, Supervision, Project administration, Methodology, Investigation, Funding acquisition, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

References

Bi, J., Bennett, K.P., 2003. A geometric approach to support vector regression. Neurocomputing 55 (1), 79–108.
Chen, J.X., Hu, Y.J., Zhu, Q., Rashid, H., Li, H.K., 2023. A novel battery health indicator and PSO-LSSVR for LiFePO4 battery SOH estimation during constant current charging. Energy 282.
Chen, L., Pan, Y., Zhang, D., 2024. Prediction of carbon emissions level in China's logistics industry based on the PSO-SVR model. Mathematics 12 (13), 1980.
Chen, L.C., Wang, S.L., Wang, Q.K., 2016. Ecosystem carbon stocks in a forest chronosequence in Hunan Province, South China. Plant Soil 409 (1–2), 217–228.
Chimner, R.A., Resh, S.C., Hribljan, J.A., Battaglia, M., Bourgeau-Chavez, L., Bowser, G., Lilleskov, E.A., 2023. Mountain wetland soil carbon stocks of Huascarán National Park, Peru. Front. Plant Sci. 14, 1048609.
Dube, T., Mutanga, O., 2015. Investigating the robustness of the new Landsat-8 operational land imager derived texture metrics in estimating plantation forest aboveground biomass in resource constrained areas. ISPRS- J. Photogramm. Remote. Sens. 108, 12–32.
Fang, H.F., Cao, J., Cai, L.H., Zhou, T., Wang, M.Q., 2021. The recognition of plastic bottle using linear multi hierarchical SVM classifier. Int. J. Fuzzy Log. Intell. Syst. 40 (6), 11509–11522.
Fu, X.H., Zheng, Q.X., Jiang, G.M., Roy, K., Huang, L., Liu, C., Li, K., Chen, H.L., Song, X.Y., Chen, J.Y., Wang, Z.X., 2023. Water quality prediction of copper-molybdenum mining-beneficiation wastewater based on the PSO-SVR model. Front. Env. Sci. Eng. 17 (8, SI).
Ganaie, M.A., Tanveer, M., Beheshti, I., 2023. Brain age prediction with improved least squares twin SVR. IEEE J. Biomed. Heal. Inf. 27 (4), 1661–1669.
GhaffariRazin, S.R., Majd, R.D., Hooshangi, N., 2023. Regional modeling and forecasting of precipitable water vapor using least square support vector regression. Adv. Space Res. 71 (11), 4725–4738.
Gu, B.J., Fang, J.W., Pan, F., Bai, Z.H., 2020. Fast clustering-based weighted support vector regression. Soft Comput. 24 (8, SI), 6101–6117.
Han, Y., Wang, M., Du, M., Guo, T., 2024. Research on deformation uniformity control of thin-walled conical aeroengine forgings based on GA-SVR. Int. J. Adv. Manuf. Technol. 131 (3–4), 1211–1222.
Hill, T.C., Ryan, E., Williams, M., 2012. The use of CO2 flux time series for parameter and carbon stock estimation in carbon cycle research. Glob. Chang. Biol. 18 (1), 179–193.
Huang, H., Ding, S., Shi, Z., 2013. Primal least squares twin support vector regression. J. Zhejiang Univ. Sci. C 14 (9), 722–732.
Huang, H.J., Wei, X.X., Zhou, Y.Q., 2022. An overview on twin support vector regression. Neurocomputing 490, 80–92.
Jia, Z., Zhang, Z., Cheng, Y., Buhebaoyin, Borjigin, S., Quan, Z., 2024. Grassland biomass spatiotemporal patterns and response to climate change in eastern Inner Mongolia based on XGBoost model estimates. Ecol. Indic. 158, 111554.
Johnson, L.K., Mahoney, M.J., Desrochers, M.L., Beier, C.M., 2023. Mapping historical forest biomass for stock-change assessments at parcel to landscape scales. For. Ecol. Manag. 546, 121348.
Karasu, S., Saraç, Z., 2020. Classification of power quality disturbances by 2D-Riesz transform, multi-objective grey wolf optimizer and machine learning methods. Digit. Signal Prog. 101, 102711.
Karasu, S., Saraç, Z., 2022. The effects on classifier performance of 2D discrete wavelet transform analysis and whale optimization algorithm for recognition of power quality disturbances. Cogn. Syst. Res. 75, 1–15.
Li, Z.X., Xie, F., Ma, J.H., Qi, Z., Wang, Y.B., 2023. Cnn-based adaptive subtraction for the removal of seismic multiples. J. Seism. Explor. 32 (2), 169–184.
Liu, Y., Fan, Y., Feng, H., Chen, R., Bian, M., Ma, Y., Yue, J., Yang, G., 2024a. Estimating potato above-ground biomass based on vegetation indices and texture features constructed from sensitive bands of UAV hyperspectral imagery. Comput. Electron. Agric. 220, 108918.
Liu, Y., Fan, Y., Yue, J., Jin, X., Ma, Y., Chen, R., Bian, M., Yang, G., Feng, H., 2024b. A model suitable for estimating above-ground biomass of potatoes at different regional levels. Comput. Electron. Agric. 222, 109081.
Liu, Z.Y., Xu, Y.K., Duan, G.F., Qiu, C., Tan, J.R., 2021. Accurate on-line support vector regression incorporated with compensated prior knowledge. Neural Comput. Appl. 33, 9005–9023.
Luo, W.X., Kim, H.S., Zhao, X., Ryu, D., Jung, I., Cho, H., Harris, N., Ghosh, S., Zhang, C., Liang, J., 2020. New forest biomass carbon stock estimates in Northeast Asia based on multisource data. Glob. Chang. Biol. 26 (12), 7045–7066.
Mirjalili, S., Lewis, A., 2016. The whale optimization algorithm. Adv. Eng. Softw. 95, 51–67.
Mohammed, M.A., Ahmed, M.A., Hacimahmud, A.V., 2023. Data-driven sustainability: leveraging big data and machine learning to build a greener future. Babylon. J. Artif. Intell. 2023, 17–23.

Mostafa Bozorgi, S., Yazdani, S., 2019. IWOA: An improved whale optimization algorithm for optimization problems. J. Comput. Des. Eng. 6 (3), 243–259.
Nguyen, H., Bui, X., Choi, Y., Lee, C.W., Armaghani, D.J., 2021. A novel combination of whale optimization algorithm and support vector machine with different kernel functions for prediction of blasting-induced fly-rock in quarry mines. Nat. Resour. Res. 30 (1), 191–207.
Ni, J., 2013. Carbon storage in Chinese terrestrial ecosystems: approaching a more accurate estimate. Clim. Change 119 (3–4), 905–917.
Pan, Y., Birdsey, R.A., Fang, J., Houghton, R., Kauppi, P.E., Kurz, W.A., et al., 2011. A large and persistent carbon sink in the world's forests. Science 333 (6045), 988–993.
Park, J.E., Nguyen, V.H., Tsai, P., Toyoda, H., Leong, J., Guy, J.E., et al., 2024. Racial and ethnic disparities in untreated patients with hepatitis C virus-related hepatocellular carcinoma but not in those with sustained virologic response. Aliment. Pharmacol. Ther. 59 (6), 742–751.
Peng, X.J., 2010. TSVR: An efficient twin support vector machine for regression. Neural Netw. 23 (3), 365–372.
Sabat-Tomala, A., Raczko, E., Zagajewski, B., 2024. Airborne hyperspectral images and machine learning algorithms for the identification of lupine invasive species in natura 2000 meadows. Remote. Sens. 16 (3), 580.
Smola, A.J., Schölkopf, B., 1998. A Tutorial on Support Vector Regression. Neuro COLT Technical Report NC-TR-98-030, Royal HollowayCollege, University of London, UK.
Valentini, R., Matteucci, G., Dolman, A.J., Schulze, E.-D., Rebmann, C., Moors, E.J., et al., 2000. Respiration as the main determinant of carbon balance in European forests. Nature 404 (6780), 861–865.
Vapnik, V.N., 1995. The Nature of Statistical Learning Theory. Springer, Berlin.
Wang, L., Gao, C., Zhao, N., Chen, X., 2021. Projection wavelet weighted twin support vector regression for OFDM system channel estimation. Artif. Intell. Rev. 54 (1), 469–489.
Wang, H., Xu, Y., Zhou, Z., 2022. Ramp loss KNN-weighted multi-class twin support vector machine. Soft Comput. 26 (14), 6591–6618.
Wang, X.N., Zhao, Y.B., Li, C., Ren, P., 2023. ProbSAP: A comprehensive and high-performance system for student academic performance prediction. Pattern Recognit. 137.
Xing, J., Xu, P., Huang, Q., Yao, S., He, J., Zhou, Y., Zheng, X., 2024. A novel adaptive dynamic weight distribution strategy applied to support vector regression based framework for dynamic load identification problems. Int. J. Crashworthiness 1–10.
Xu, Y.T., Wang, L.S., 2012. A weighted twin support vector regression. Knowl.- Based Syst. 33, 92–101.
Xu, Y.T., Wang, L.S., 2014. K-nearest neighbor-based weighted twin support vector regression. Appl. Intell. 41 (1), 299–309.
Zhang, Z.C., Ding, S.F., Sun, Y.T., 2021. MBSVR: Multiple birth support vector regression. Inform. Sci. 552, 65–79.
Zhang, S.C., Li, J.Y., 2023. KNN classification with one-step computation. IEEE Trans. Knowl. Data Eng. 35 (3), 2711–2723.
Zhang, X., Wang, T., Lai, Z., 2024. A feature-weighted support vector regression machine based on Hilbert–Schmidt independence criterion least absolute shrinkage and selection operator. Information 15 (10), 639.
Zhao, Y.P., Zhao, J., Zhao, M., 2013. Twin least squares support vector regression. Neurocomputing 118, 225–236.
Zhu, W., Li, Y., Luan, K., Qiu, Z., He, N., Zhu, X., Zou, Z., 2024. Forest canopy height retrieval and analysis using random forest model with multi-source remote sensing integration. Sustainability 16 (5), 1735.